Motor Imagery EEG Classification Using Fine-Tuned Deep Convolutional EfficientNetB0 Model

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Abstract—**This work proposes a new way to apply the deep convolutional EfficientNetB0 model for the classification to learn various electroencephalogram (EEG) signal properties on BCI competition IV dataset 2b. Several deep convolutional neural networks (DCNN)-based techniques have been applied to enhance the accuracy of motor imagery-based brain-computer interfaces (BCIs). The time-varying nature of various frequency bands makes extracting informative features from EEG signals difficult, causing the loss of DCNN classification accuracy. The EfficientNetB0 baseline model's feature extraction capability is used to overcome these limitations by first transforming EEG 1D signals into 2D images using the feature extraction technique of the Short-Time Fourier Transform (STFT) algorithm to train and evaluate the EfficientNetB0 model. The transfer learning approach is used to expand the initial feature sets for efficient model training in a short period. After learning from the dataset, the entire model is retrained and fine-tuned to work with the proposed layers. Our evaluated results demonstrated that the highest average Accuracy, Precision, Recall,F1-Score and MCC with the STFT method is 86.46%,88.2%, 91.2%,89.78% and 0.815 respectively over 10 epochs. According to the results, the proposed methodology outperforms other state-of-the-art DCNN models for feature extraction and classification of two-class motor imagery, namely right-hand and left-hand movements for the given dataset.**

Keywords—Motor Imagery (MI), EfficientNetB0, Deep Convolutional Neural Networks (DCNNs), Fine-Tuning, Transfer Learning, Short Time Fourier Transform (STFT)

I. Introduction

Brain-Computer interfaces (BCI) are emerging technologies that enable users to operate computers or digital devices directly without human interaction. The BCI- based system characteristically records the brain signals created by the user and uses feature extraction, classification, and pre-processing techniques to ascertain the user's intention. Electroencephalography (EEG) is the most frequently used approach for recording brain activity. It allows input signals to feed into BCI-based systems because of its low cost and non-invasive nature among the various approaches. A BCI that produces sensorimotor rhythms and targets the motor cortex area of the brain is called Motor Imagery BCI [1].

In the recent past, Convolutional Neural Networks (CNNs) approaches are mostly utilized by researchers to enhance the performance of motor imagery (MI)-based BCI systems [2]. A CNN optimization technique for motor imagery (MI)-based EEG classification-related tasks is presented [3]. In this technique, two CNN architectures are suggested and tested for various configurations and characteristics. The authors highlighted that CNN-based methods outperform traditional approaches, including naive Bayes (NB) and support vector machines (SVMs) algorithms based on their experimental results.

CNN has recently successfully solved real-world problems like image classification, recognition, localization, and detection. The use of CNNs that are more extensive, deep, and the complex has become a standard in the domain of computer vision. CNN models such as CapsNet [8] have made significant advances and provide vulnerabilities such as increasing neural network width with separable convolutions, additional layers, and dropout layers from increased model volume for regularization [4]. As the complexity of the previously built CNN increased, they evolved into Deep Convolutional Neural Networks (DCNNs). However, because of the architectural complexity and data requirements, duplicating DCNNs for particular purposes requires substantial computational power. To overcome these limitations, a new approach known as transfer learning emerged [4]. The problem of large-scale DCNN reusability is solved using ImageNet's pre-trained weights through the transfer learning approach, which contains essential image recognition features to train other models. The transfer learning approach enables DCNNs to perform particular tasks further than their novel purpose without using more computational power through fine-tuning techniques [5]. The transformer model's weights are typically used as the starting point for the new task in the transfer learning approach, with the final layers being retrained on the new dataset. This approach is used because it can be more efficient and requires less data than training a model from scratch. Additionally, the pre-trained model has already learned useful features leveraged for the new task.

In this study, a DCNN through a transformer-based EfficeintNetB0 model is applied to classify two-class motor imagery EEG signals using ImageNet's pre-trained weights that automatically extract various features from the 2D images to be learned. For this purpose, motor imagery 1D signals are first transformed into 2D images using the STFT. The transformed 2D images are used for training and testing the EfficientNetB0-based model. The public dataset BCI Competition IV dataset 2b [6] is used for training and evaluating the proposed method, and an accuracy of 86.46% is obtained for the STFT method. The proposed methodology also outperforms as compared with other DCNN-based state-of-the-art classification methods that have been commonly used to classify EEG signals. The proposed methodology is the first attempt to use EfficientNetB0 for the MI-EEG signal classification tasks.

The rest of the paper is organized as follows: Section II describes the literature review, whereas section III presents the proposed methodology. The implementation of the proposed methodology is done in section IV. Section V highlights the main results of this work and comparative analysis. Furthermore, sections VI and VII present the discussion, a conclusion and future perspectives.

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II. Literature Review

In this section, a detailed literature review is carried out to ascertain the current state of research on the classification of MI tasks.

A. Related Work

A deep learning (DL) approach for classifying EEG signals through CNN and stacked autoencoders (SAE) technique is presented [7]. The authors used the Time-Frequency Analysis (TFA) for features extraction on BCI competition IV dataset 2b and obtained 54.7% accuracy. Ha and Jeong [8] Presented the capsule network (CapsNet) approach for EEG signal classification on the data set BCI competition IV dataset 2b. The authors utilized the Short-time Fourier transform (STFT) technique to extract features from 2D images. The accuracy achieved is 78.4%, and the results also outperformed compared to other classifiers such as SVM, CNN, and LDA. Lu et al. [9] proposed a novel scheme based on the restricted Boltzmann machine (RBM). Fast Fourier transform (FFT) and wavelets decomposition (WD) approaches are applied for feature extraction purposes to the frequential deep belief network (FDBN). The authors got 84% accuracy. Chatterjee et al. [10] classified motor imagery signals using statistical and wavelet-based features. This study made use of the BCI competition II dataset. The energy and entropy of all wavelets are calculated using a fourth-order Daubechies wavelet. Welch's power spectral density (PSD) calculates the average power for mu and beta rhythms. For the classification of signals, SVM and MLP models are used. The statistical characteristics resulted in a performance accuracy of 78.57% and a ROC of 0.79. Likewise, wavelet decomposition produced an accuracy of 85% and a ROC of 0.85. Chaudhari and Galiyawala [11] proposed a backtracking search optimization strategy for the motor imagery of the upper limbs classification. PSD features are calculated on the BCI competition III dataset. The accuracy acquired by the authors for Subject 1 is 80.32%, Subject 2 is 66.03%, and Subject 3 is 59.34%. Z. Chen et al. [12] proposed a framework named IS-CBAM-CNN to handle the temporal localization of excitation occurrence, non-stationary nature, and frequency band distribution characteristics of the MI-EEG signal on the BCI competition IV dataset 2b. The framework achieved an accuracy of 79.6%.

B. Research Gap

For MI tasks, the following works made significant contributions. However, as mentioned in Section I, recent developments in DCNN models also significantly impact the classification of motor Imagery (MI)-based tasks. As DCNNs have some limitations. First, DCNNs only perform properly if the dataset is highly variable and distorted, even for the same person. Second, DCNNs can learn limited spatial features due to pooling that results in performance degradation of DCNNs in terms of accuracy metrics [9]. A recent study presented that a new DCNN, like the EfficientNetB0 model, could better classify various medical images, like chest x-rays [13]. Their work demonstrates that the EfficientNetB0 model could also achieve better results in other areas of medical imaging. The EfficientNetB0 model can scale each feature matrix dimension using a fixed set of scaling factors in the form of parameter size and precision from 0 to 7. This model also outperforms other DCNN-based state-of-the-art classification models trained on ImageNet's dataset, demonstrating its effectiveness in image classification tasks [13].

This research mainly contributes to the first attempt using the EfficientNetB0 model on motor imagery classification-related tasks. The proposed model incorporates fine-tuned final layer modified by freezing layers to make it trainable to classify two classes of motor imagery: right and left-hand movements. The proposed methodology aims to create a model that consumes little disk space, easy to deployable, and is modifiable as needed. In addition, the proposed methodology outperformed other DCNN-based state-of-the-art classification models that performed the motor imagery classification-related tasks.

III. Proposed Methodology

This section describes the methodology used to classify motor imagery signals. This methodology is divided into two phases: pre-processing and classification. In the pre-processing phase, band-pass filtering and EEG segmentation of raw multi-channel EEG signals is performed. After that, the signals are converted into the frequency domain through the Short-Time Fourier transform (STFT), frequency band separation, and feature extraction. During the classification phase, the extracted features are utilized for training and testing the EfficientNetB0 model. The proposed methodology is shown in "Fig. 1".

A. Data Set

BCI competition IV dataset 2b [6] is used to validate the proposed methodology. The EEG dataset consisted of nine subjects and 4480 data samples. Each subject attended five sessions; the first two sessions recorded training data without visual feedback, and the next three sessions have used to record training data with feedback. The training data without visual feedback is used in this study. Three channels are considered, namely, along C3, Cz, and C4, and recordings are made at a sampling rate of 250 Hz. Since motor imagery tasks do not reveal the effect of Cz, so Cz channel is ignored during data processing. There are 6 runs in each recording, with 10 trails for each class. A short beep serves as a warning tone, and the subjects are shown a fixation cross before actual imagination. After this fixation cross, a visual cue that showed an arrow pointing either right or left, depending on the class, is shown for 1.25 seconds. The subject has to imagine the hand movement for 4 seconds after this cue. After each image, the subject is asked to take a short break of approximately 1.5 seconds [6].

B. Band Pass Filtering

Band-pass filtering is a common preprocessing step in EEG signal analysis.Before being processed for motor imagery classification, the recorded EEG signals from subjects must be noise free. For this, the Notch band-pass filter, with a lower cutoff frequency of 0.1 Hz and a higher cutoff frequency of 50 Hz, is used for band-pass filtering of EEG signals.

C. EEG Segmentation

The EEG signals are non-stationary because of their time-varying nature of statistical features. This problem is solved by dividing a long sequence of EEG signals into short, pseudo-stationary segments with similar statistical time and frequency features. For this, EEG signals are split into 4 seconds segments with overlapping frames where actual movement occurred.

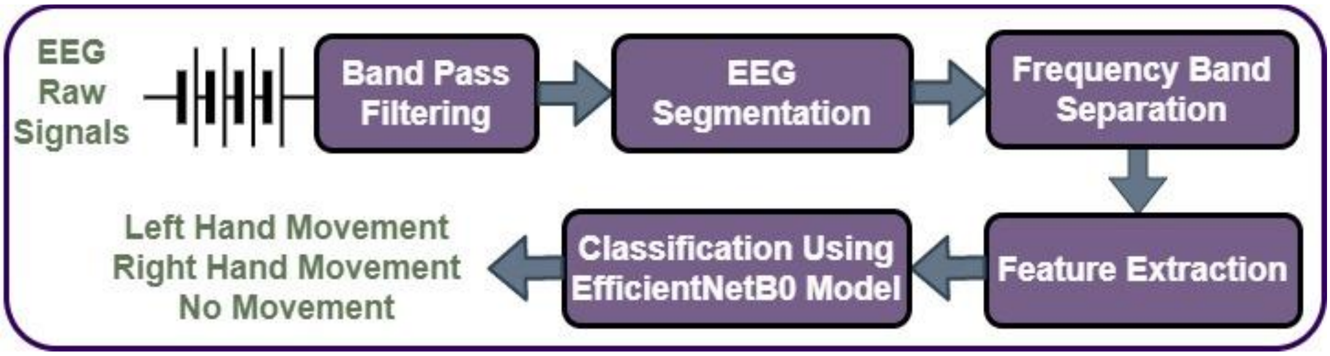

Fig. 1. Proposed Methodology.

D. *Frequency Band Separation*

There is a need to preserve the activation patterns generated by motor imagery tasks that are primarily related to mu (μ) and beta (β), delta (δ), and theta (θ) bands [6]. For this, the μ (8–15 Hz) and β (15–30 Hz) bands of EEG data are explicitly utilized rather than the spectral contents of entire bands. The shifting characteristics of the μ and β bands frequently serve as predictors for motor activities. After extracting the specified events (Left/Right) and channels (C3, Cz, and C4) by passing the frequency band through the slicing of the frequency axis by a 4:15 ratio, the frequency band-separated EEG signals are obtained as shown in "Fig. 2".

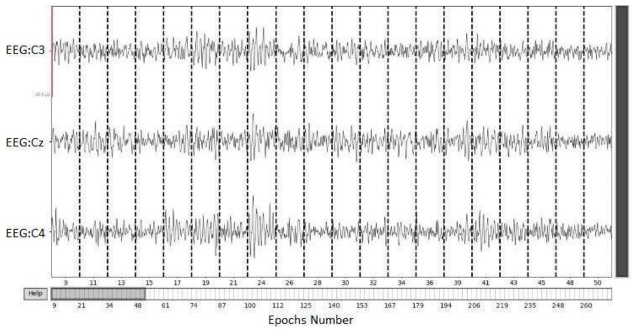

Fig. 2. Frequency band-separated EEG signals.

E. *Feature Extraction*

Many different time-frequency domain analysis techniques have been used to extract features effectively for motor imagery tasks. To extract the features from the 1D signal, the Short-Time Fourier transform (STFT) is used in the proposed methodology.

1) *Short-Time Fourier Transform (STFT)*

STFT makes a 2D image from a 1D signal. After EEG signal segmentation in Section III-C, it determines frequency changes at a specific time point [8]. STFT is applied on time series data with a 4 seconds interval corresponding to the 4480 data samples for 250 Hz where movement occurred. STFT is carried out with a window size of 140 samples and an overlap size of 100 samples. The 1D signals for the electrodes C3, and C4, connected to motor imagination tasks, are subjected to STFT. The vertical concatenation of the resulting 2D images of both channels (μ and β band) for the C3 and C4 channels are obtained. The Cz channel is ignored because it does not provide information about brain activity patterns. The feature extracted EEG signals from ID signals is shown in "Fig. 3".

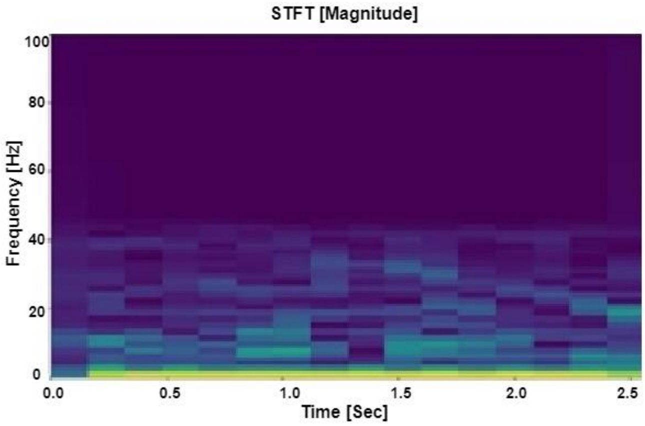

Fig. 3. Feature extracted EEG signals through STFT.

F. *Classification*

The EEG signals are extremely susceptible to external noise due to low amplitude. As a result, even the artifacts, such as eye blinks and muscle movements, can negatively impact the EEG signals. In addition, subjects in some trials cannot complete MI tasks because they are distracted by the machine. These pieces of evidence change from trial to trial [6]. The brain's activity varies subject to subject, resulting in varying activation patterns. Therefore, there is a need for a classifier that can address all these issues.

1) *EfficientNetB0 Model*

"Fig. 4" presents the most recent baseline model for EfficientNetB0. It takes an input image with dimensions 224 x 224 x 3. The model then uses multiple convolutional layers (Conv), a mobile inverse bottleneck Conv (MBConv), and a 3x3 receptive array to extract features from each layer. The EfficientNetB0 model can scale each dimension using a fixed set of balanced depth, width, resolution, and scaling factors, resulting in a scalable, accurate, and easy-to-implement compared to other DCNNs [14].

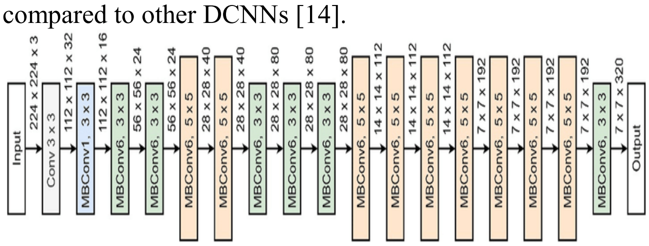

Fig. 4. EfficientNetB0 Baseline Model Architecture [14].

IV. Implementation

For the implementation of the model, the dataset [6] required image resizing because the EfficientNetB0 model requires an image with a dimension size of 224×224×3 as an input. For this, a 224 × 224 image is zero-padded with the pad of (3, 3) after cropping the images from the dataset.

A. Proposed Layers

The new final layers of the EfficientNetB0 baseline model are proposed to activate new weights from it to use the EfficientNetB0 model to learn from prepared data through fine-tuning techniques. The Global Average Pool 2D (GAP2D), two fully connected Dense layers, a Dropout layer, a Normalization, and a Softmax classifier are proposed as the EfficientNetB0 model's final layers are shown in "Fig. 5".

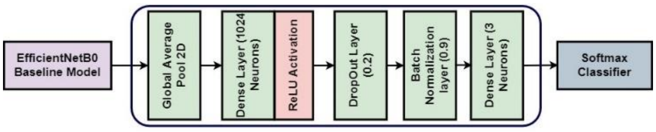

Fig. 5. Proposed Final layers for EfficientNetB0 Model.

The GAP2D layer is added to reduce the number of parameters from the base model by rescaling the width, height, and depth of the incoming tensor into 1×1×3 dimensions, respectively. A GAP2D layer acts as a pooling layer to prevent severe overfitting from the complex feature handling. Using this technique, the enormous flow of features is controllable that went to the dense layer, causing overwhelming the classifier. The GAP2D reduces the feature map portions during the technique, maintains the complex patterns necessary for image recognition, and averages all features received from the 2D images [15]. Before making predictions about the outcome, the previous GAP2D feature set is mapped with 1024 hidden neurons of the dense layer linked to an additional layer of 3 neurons demonstrating the 3 labels (C3, Cz, and C4). Each feature map received a new set of weights and biases through this approach, which generated a linear probability among all features. In the hidden layers of the 1024-dense layer, rectified linear units (ReLUs) activation function is used to set all negative input values in the layer before it to zero to provide nonlinearity and speed up training [16]. The input values are normalized using batch normalization with a momentum of 0.9. The dropout layer of 0.2 is added as a regularization method to prevent overfitting of the model. It gives the probability of dropping the number of neural nodes in the layer. The softmax classifier limits the input to the number of classes in a fully connected layer. A classifier can only use the softmax function when the classes are mutually exclusive [17]. Softmax is computed as follows:

$$\sigma(x_j) = \frac{e^{xj}}{\sum_i e^{xj}} \tag{1}$$

B. Fine Tuning and Transfer Learning Approach

"Fig. 6" shows the fine-tuning of the proposed model. First, the transfer learning approach is used to enhance the size of the training parameters. The base model instantly utilized ImageNet's pre-initialized weights, enhancing image recognition. For an image classification task, shapes, edges, and other essential components are detected with the assistance of features in ImageNet's weight. This method sped up the process with less effort concerning initialized weights. The pre-trained EfficientNetB0 baseline model has 1000 different classes and more than 14 million images [18]. The current weights and baseline architecture of the EfficientNetB0 are not used directly for motor imagery classification, so fine-tuning is essential in this situation [19]. Therefore, the base model's preliminary layers are frozen, and then fine-tuning is applied to the proposed final layers of the EfficientNetB0 model using the dataset. This approach allowed us to keep ImageNet's features intact in the feature extraction layer from 2D images and protect them from overfitting during the model's training. After retraining the entire network using ImageNet's data set and weights, the extractor and the proposed layers are trained before creating the final model. The final model is then validated against the validation data.

C. Hyperparameter and Loss Function

A deep learning (DL) model's evaluation can be assessed in terms of accuracy, performance metrics, and loss function [20]. For this, the mean squared error (MSE) loss function is used in this study because there is a need for pixel-by-pixel information to locate the activity patterns of left and right-hand movements. MSE is computed using

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - y_i^{'})^2 \tag{2}$$

Adam optimizer is used at low learning rate (LR) of 0.001 with batch size 32 to reduce loss during training. Adam is an adaptive gradient descent optimizer that accelerates weight descent towards local minima. It is easy to implement, efficient in memory consumption, and has fast training time compared to other optimizers such as stochastic gradient descent (SGD) and RMSProp in medical image analysis [21].

V. Validation

This section describes the validation of the proposed EfficientNetB0 model and the comparative analysis concerning other DCNN-based state-of-the-art models.

A. Results

The EfficientNetB0 model is evaluated on the BCI competition IV Dataset 2b. Accuracy, Recall, Precision, Matthews's correlation coefficient (MCC), F1-Score metrics and ROC curve plot are used for performance evaluation. For this, the dataset is divided into percentage-split so that 90% (4032 samples) are used for training and 10% (448 samples) for testing the model. Signals are extracted against 10 epochs of 4 seconds corresponding to movements followed by STFT with a batch size of 32 is used for the training of the dataset utilized in one iteration, a learning rate of 0.001, and the Adam optimizer by taking into consideration the exponentially weighted average of the gradients. The proposed model got a training accuracy of 87.1% and a validation accuracy of 86.5% with the STFT method. The performance evaluation results are listed in Table I.

TABLE I. Performance Evaluation Metrics Results

Metrics	Score
Training Accuracy %	87.1
Validation Accuracy %	86.5
Precision %	88.2
Recall %	91.2
F1-Score %	89.78
MCC	0.815

The model accuracy and model loss plots concerning STFT for the proposed EfficientNetB0 model are shown in "Fig. 7" and "Fig. 8", respectively, with accuracy and loss measures on the y-axis and the number of epochs measured on the x-axis.

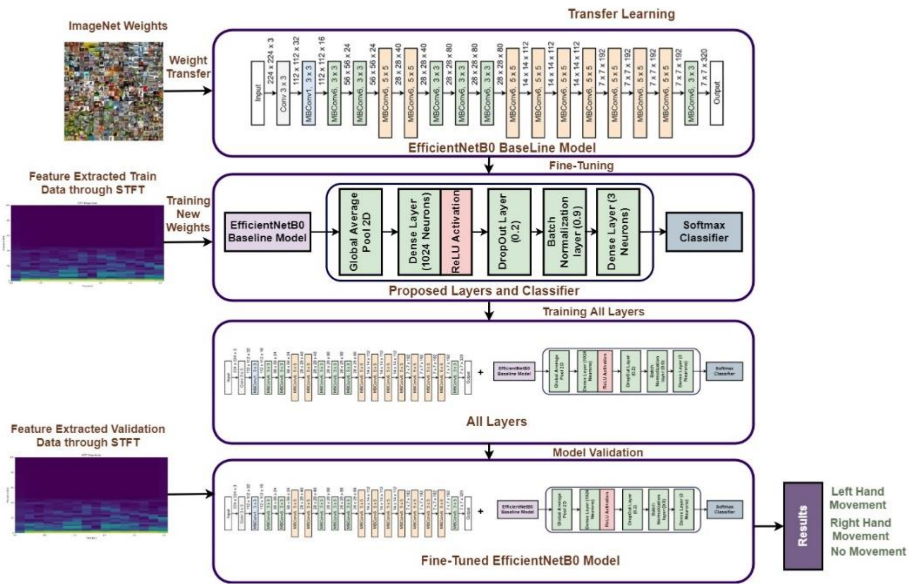

Fig. 6. Fine-tuning and Transfer learning process of the EfficientNetB0 model on BCI competition IV Dataset 2b.

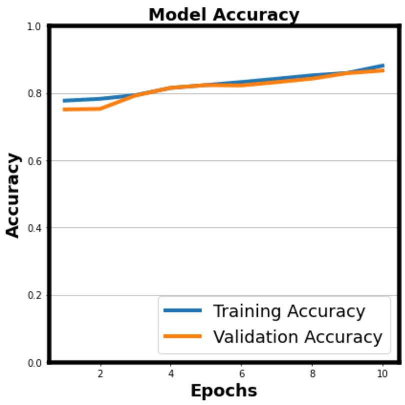

Fig. 7. Model Accuray for STFT.

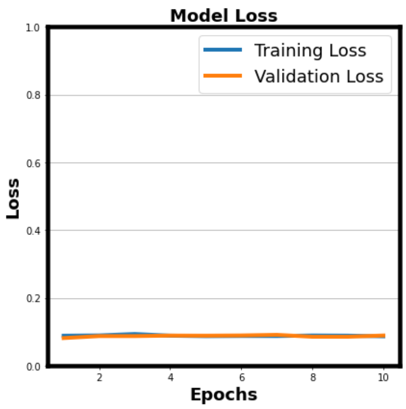

Fig. 8. Model Loss for STFT.

ROC (Receiver Operating Characteristic) curve and AUC (Area Under the Curve) are also used to evaluate the performance of different configurations of EfficientNetB0. The ROC curve is a graph that plots the true positive rate (TPR) against the false positive rate (FPR) at different classification thresholds. The AUC for the EfficientNetB0 model is 0.934 at threshold of 0.25 means that the model made perfect predictions.ROC curve plot with AUC value concerning STFT for the proposed EfficientNetB0 model are shown in "Fig. 9".

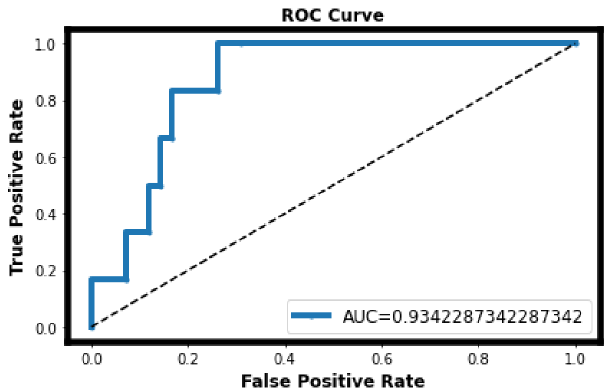

Fig. 9. ROC Curve for STFT.

B. *Comparative Analysis*

Table II compares the proposed methodology's accuracy recent FDBN [9], CapsNet [8] and IS-CBAM-CNN [12] classification methods. For this, average accuracy (mean) concerning standard deviation are used to calculate 9 subjects accuracy separately. The results showed the highest accuracy of 86.46 % ±6.17, among other DCNN classification techniques applied to MI-related tasks on BCI competition IV dataset 2b [6].

TABLE II. COMPARAISON OF EFFICIENTNETB0 AND OTHER STATE-OF-THE-ART METHODS

Sub ject	EfficientNet B0 +STFT% (Proposed)	(FDBN) + FFT% [9]	CapsNet+ STFT % [8]	IS-CBAM-CNN[12]
1	87.5 ± 3.75	81.0	78.75	80.3 ± 1.5
2	85.42 ± 6.88	65.0	55.71	75.0 ±1.8
3	89.58 ± 6.87	66.0	55	67.7 ± 2.6
4	84.62 ± 7.12	98.0	95.93	95.4 ± 0.6
5	88.46 ± 6.73	93.0	83.12	88.3 ± 1.5
6	85.42 ± 6.46	88.0	83.43	80.0 ± 1.7
7	87.5 ± 5.42	82.0	75.62	73.7 ± 2.2
8	82.14 ± 4.64	94.0	91.25	77.4 ± 2.0
9	87.5 ± 7.9	91.0	87.18	78.6 ± 2.1
Ave rage	**86.46 ± 6.17**	84.0	78.44	79.6 ± 1.8

VI. DISCUSSION

Our evaluation results show that the fine-tuned EfficientNetB0 model performs well as it does not require extensive data preprocessing, data augmentation, and even long training epochs for the model's training. However, since there is no discernible pattern of motor imagery signals at Cz, so Cz channel is ignored, and only the C3 and C4 channels of the electrodes are used. Efficient data preprocessing is performed on time series signals that are first extracted into 2D images using STFT to avoid spatial relationships during feather extraction, and fine-tuning is performed on the EfficientNetB0-based model to combine with the proposed final layers with a less rigorous choice of hyperparameter standards for model training against 10 epochs to achieve this. The rectified linear units (ReLUs) activation function is used, which maps the input to 0 if it is below to prevent the model from vanishing gradient problems. To prevent the model from overfitting, a drop-out layer of 0.2 is used as a regularization technique in the proposed model. That fine-tuned and pre-trained model efficiently classifies the difference between left and right-hand movements in EEG signals by detecting subtle features. According to Table II, the proposed methodology performs better than other DCNN approaches.

VII. CONCLUSION

In this study, an EfficientNetB0-based model is trained with a transfer learning approach to use pre-trained ImageNet's weights, and fine-tuned final layers of the EfficientNetB0 model for classifying two-class motor imagery (MI)-based EEG signals are presented. A set of transformed 2D images through STFT are taken from raw EEG signals of the BCI competition IV 2b dataset that is used as the proposed method's input after the normalization of the input image to a 224×224×3 dimension to validate the efficiency of the selected baseline model. A dataset comprised of three electrodes is used to evaluate the proposed method. The model is trained on 10 epochs, and the accuracy rate of 86.46% is achieved with the STFT method. The EfficientNetB0 model also outperformed the other state-of-the-art classification models like FDBN and CapsNet model concerning accuracy. These results led to the conclusion that the transfer learning-based fine-tuned EfficientNetB0 model could effectively classify two-class motor imagery EEG signals and has the capability of exceptional results despite the deficit of complicated image pre-processing, image intensification, and difficult image optimization techniques. The proposed methodology may benefit from further improvements, i.e., by increasing the electrodes used to generate the input images aligned to visualize the elicited patterns efficiently. Another proposed improvement could be to add more layers in the final layer to fine-tune the EfficientNetB0 model more effectively may be another potential improvement area.

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